Worldline | UNST Politehnica Bucharest | ACS | 3iL Engineers

POLITEHNICA UNIVERSITY OF BUCHAREST

Faculty of Automatic Control and Computers — MSc in Artificial Intelligence

3iL ENGINEERS — Double degree

Hybrid-AI Log Diagnostics

DRAIN + LogBERT + SLM Pipeline for Anomaly Detection and MTTR Reduction

Double-degree MSc thesis in AI — UPB Bucharest / 3iL Engineers

---

Thesis presented by

SYLLA Abdou Lahat

Academic year:	2025/2026
Host company:	Worldline — Trading & Financial Information (TFI) division
Team:	Client Delivery 2
Company supervisors:	Ludovic Charvoz, Doreen Lucky Tchoupe Soh

Protocol:	normal-only training, validation selection, locked test
Data:	full HDFS, 575,061 sessions, 11,175,629 events
Reference:	Guo et al., LogBERT, IJCNN 2021
Selected model:	V2 m=0.75, g=8, r=0.04

Working document — figures from the Kaggle GPU T4 run (official LogBERT repository, commit `a845e61`).

Contents

1 Hybrid-AI Log Diagnostics: DRAIN + LogBERT + SLM Pipeline 3

1.1 General Introduction . . . 3

1.2 Host Company Presentation . . . 3

1.3 Reflection Background . . . 4

1.3.1 Operational Context and Log Management Reality . . . 4

1.3.2 Project Timeline and Origins . . . 4

1.3.3 Problem Statement . . . 4

1.4 State of the Art . . . 5

1.4.1 Threshold Supervision and Alert Fatigue . . . 5

1.4.2 The Monolithic-LLM Dead End . . . 5

1.4.3 DRAIN and LogBERT . . . 5

1.4.4 Comparison with Existing Methods (LogBERT Paper SOTA) . . . 5

1.5 Pipeline Architecture . . . 7

1.6 Experimental Protocol . . . 7

1.7 Training: Train / Validation Curves . . . 8

1.8 Locked-Test Results . . . 8

1.9 BGL Generalisation and Multi-Seed Rigor (V3) . . . 10

1.10 Inference and Enterprise Applicability . . . 12

1.10.1 Bundle Inference Contract . . . 12

1.10.2 Worldline (TFI) Integration . . . 13

1.10.3 Enterprise Use Case and Causal-Diagnostic Example (JSON) . . . 13

1.10.4 FinOps and Sovereignty 14
1.10.5 Risks, Limitations, Acceptability 15
1.11 Conclusion 15

Chapter 1

Hybrid-AI Log Diagnostics: DRAIN + LogBERT + SLM Pipeline

1.1 General Introduction

This report addresses a single enterprise problem: diagnosing incidents buried in massive banking log streams quickly and at controlled cost, without moving data outside the private network. Conducted as an apprenticeship at Worldline (Client Delivery 2 team, TFI division) within the AI Master's double degree (Politehnica University of Bucharest / 3iL Engineers), it combines operational field experience with applied research: a hybrid pipeline in which DRAIN structures logs, LogBERT detects anomalies statistically, and a constrained SLM explains them for on-call engineers. The outline follows this thread: company presentation, problem statement, state of the art, architecture and protocol, training, locked-test results, inference and production applicability, then conclusion.

1.2 Host Company Presentation

Worldline, a European payment services provider (4th worldwide), operates from Paris La Défense in over 50 countries with more than 18,000 employees. My apprenticeship took place in the **Client Delivery 2** team of the **Trading & Financial Information (TFI)** division within Financial Services: specialised hosting and operational maintenance of critical infrastructures for banking clients (Copartis, CommerzBank, Boursorama), on the in-house private cloud and on-premise estates.

Only the Copartis context is used in this thesis, as motivation for the AI topic: ∼172 million log lines per month excluding INFO/WARN (66 logs/s on average, peaks of 2,506 logs/s over 30 days) and a cascade incident on the CommerzBank CRM (∼4,600 lines, three

components) rooted in the infrastructure layer (missing Oracle library, memory saturation, database host resolution failure). This need — qualifying an incident quickly in massive streams without exporting data outside the private network — motivates the DRAIN + LogBERT + SLM pipeline studied in the double-degree AI MSc (UPB Bucharest / 3iL).

1.3 Reflection Background

1.3.1 Operational Context and Log Management Reality

Day-to-day operation of the scope (presented above) relies on threshold rules and on-call expertise. Two lived limitations motivate this work: (i) thresholds miss slow degradations and drown on-call engineers in cascades of correlated alerts; (ii) facing a cascade with no obvious cause, attributing the incident to the platform layer or to client code requires manually sifting thousands of lines — as in the CRM incident, resolved by isolating three platform causes through assisted analysis (internal LibreChat). Hence the goal: moving from manual triage to statistical detection coupled with constrained explanation.

1.3.2 Project Timeline and Origins

MSc semester in AI at the Politehnica University of Bucharest (S1 2025/2026): attention [1], VAEs [2], BERT [3]. Back in the company on 16 February 2026: the initial monolithic project (Ludovic Charvoz, Doreen Lucky Tchoupe Soh) had been abandoned over volume, interfacing, confidentiality and cost roadblocks. Hence the proposal of a frugal, modular and sovereign architecture, the subject of this work.

1.3.3 Problem Statement

> *Faced with exploding log volumes and cloud cost constraints, how does hybridising DRAIN + LogBERT + SLM significantly reduce MTTR while guaranteeing FinOps sustainability?*

Engineering requirements: reproducibility, test-leak-free protocol, traceable decisions, bounded cost, data confined to the private network.

1.4 State of the Art

1.4.1 Threshold Supervision and Alert Fatigue

Threshold rules (e.g. `LBHealthCheckServerDown`: HAProxy backend DOWN for > 2 min) catch hard failures but miss slow degradations and produce avalanches of correlated alerts that desensitise on-call teams [17]. Classical parsing (Drain [4], Spell, IPLoM) and detection (DeepLog [9], LogAnomaly [15]) approaches are surveyed by Zhu et al. [7] and the LogHub benchmark [8].

1.4.2 The Monolithic-LLM Dead End

Context windows saturated by continuous streams, quadratic attention cost, signal dilution and factual hallucinations unacceptable in banking [18]. Small instruction-tuned models (Qwen2.5 [19], Llama-3 [20]) only change the equation if they act downstream of strict statistical filtering.

1.4.3 DRAIN and LogBERT

DRAIN [4] routes each log by length and prefix into a fixed-depth tree and extracts the template by masking variables. Figure 1.1 illustrates this raw-log-to-template passage: variable parts (identifiers, addresses, timestamps) are replaced with wildcards, reducing each message to its structural signature.

LogBERT [5] trains a Transformer on log-key sequences with MLKP (masked log-key prediction, ratio m) and VHM (compactness of normals). Paper-style scoring: for each masked position, an error is counted when the observed key falls outside the top-g candidates; a session is anomalous when its error proportion exceeds r. The same m is used in training and inference.

1.4.4 Comparison with Existing Methods (LogBERT Paper SOTA)

The LogBERT paper [5] evaluates six baselines from three families on HDFS, BGL and Thunderbird: count-vector + classical detector methods (**PCA** [10], **One-Class SVM** [11], **Isolation Forest** [12], **LogCluster** [13]) and recurrent networks (**DeepLog** [9], **LogAnomaly** [15]). All use Drain [4] for parsing; PCA/OCSVM/iForest/LogCluster are evaluated via Loglizer [14]. Table 1.1 reproduces the paper's HDFS figures (Table 2, test with 553,366 normal / 10,647 anomalous sequences).

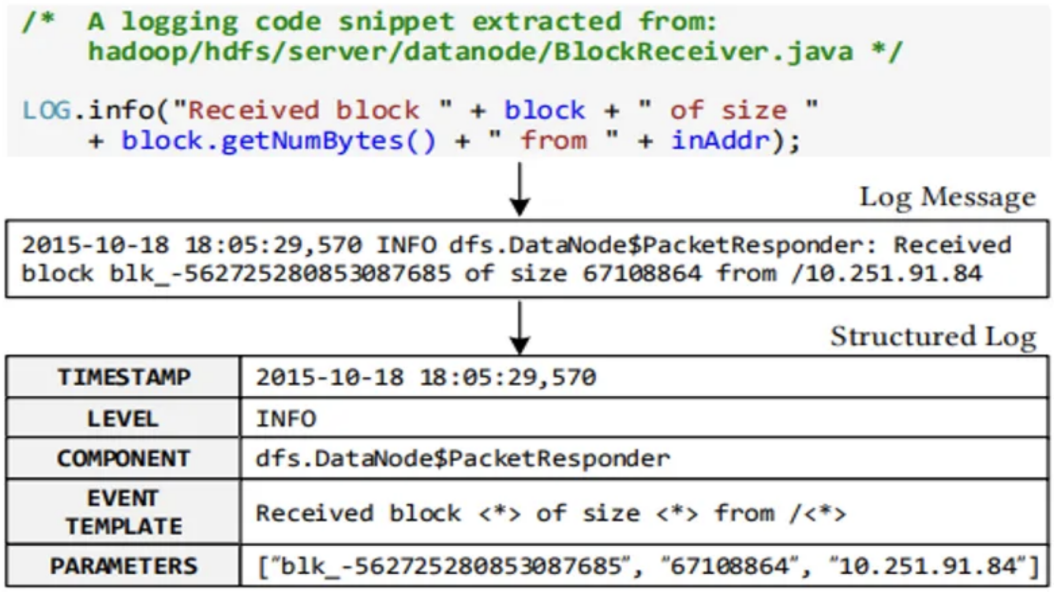


Figure 1.1: DRAIN parsing: from raw log to structured template (variables masked, fixed skeleton preserved).

Method	Precision	Recall	F1
PCA	0.0589	1.0000	0.1112
Isolation Forest	0.5360	0.6941	0.6049
OCSVM	0.0254	1.0000	0.0495
LogCluster	0.9926	0.3708	0.5399
DeepLog	0.8844	0.6949	0.7734
LogAnomaly	0.9415	0.4047	0.5619
LogBERT (paper)	**0.8702**	**0.7810**	**0.8232**

Table 1.1: State of the art on HDFS after Guo et al. (2021), Table 2. Source: `2103.04475v1.pdf`.

Reading: count-based methods sacrifice temporal order (PCA/OCSVM at recall 1.0 but near-zero precision); LogCluster rises in precision but misses half the anomalies; DeepLog and LogAnomaly confirm the value of sequential modelling, with the best classical F1 at 0.7734 (DeepLog). LogBERT (0.8232) outperforms them with superior precision/recall balance. The paper's ablation (Table 3) explains this gain: MLKP alone yields 0.7754 / 0.7865 / 0.7809, VHM alone 0.0243 / 0.3917 / 0.0458, and combined MLKP+VHM 0.8702 / 0.7810 / 0.8232 — VHM contributes +4.2 pts of F1 on HDFS with short sequences (average length 19), where its contribution is marginal on BGL/Thunderbird with long sequences. Our V1/V2 protocol reuses this foundation identically (same top-g/r scoring,

same m in training and inference) before testing its sensitivity to m.

Paper baseline (Guo et al., 2021, HDFS): Precision 87.02%, Recall 78.10%, F1 82.32%. Indicative, non-paired reference: different split, official predictor threshold tuned on test, 4 layers in the code vs. 2 described. Repository: https://github.com/HelenGuohx/LogBERT.git, commit `a845e61` [6].

1.5 Pipeline Architecture

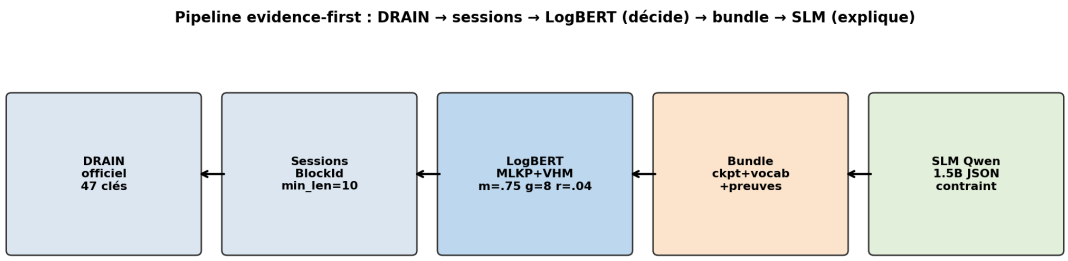


Figure 1.2: Evidence-first funnel architecture. LogBERT decides; the deterministic policy filters; the SLM explains.

Five stages: (1) official DRAIN parsing → 47 keys; (2) sessionisation by `BlockId`, `min_len`=10; (3) LogBERT MLKP+VHM; (4) AIOps bundle (`best_bert.pth`, `vocab.pkl`, Drain mapping, `inference.py`, manifest) with an API over `EventId` sequences; (5) Qwen2.5-1.5B-Instruct SLM in constrained JSON (engineer summary, LogBERT reading, probable causal chain, uncertainties, P1–P3 actions, cited events). The SLM never re-decides; any free-form action is forbidden (finite catalogue + deterministic parameter injection), with a single reading direction: the further right in Figure 1.2, the less data and the more qualified its content.

1.6 Experimental Protocol

Kaggle GPU T4. HDFS_v1 data (Zenodo 8196385, MD5 `76a24b4d9a6164d543fb275f89773260`), logpai/loglizer labels. **Normal-only training**: 4,855 sessions (seed-20 shuffle from the official script). **Validation**: 55,336 normal / 2,139 anomalous. **Locked test**: 498,032 / 8,508 (1.68% prevalence), disjoint `BlockIds`, 575,061 sessions and 11,175,629 events in total (median 19, p95 28). Vocabulary derived from training only. Selection of m, g, r on validation (projection at reference prevalence); test metrics at real prevalence + 95% Wilson intervals.

V2 grid: $m \in \{0.50, 0.65, 0.75\}$, $g \in \{1, \ldots, 15\}$, $r \in [0, 0.80]$ step 0.01, single seed. V1 = official baseline m=0.65, g=6, r=0. Compatibility fixes with no model change: `torch._six`, `np.float/int/Inf`, ragged `dtype=object`, pandas 2.2.3 / NumPy 2.x.

1.7 Training: Train / Validation Curves

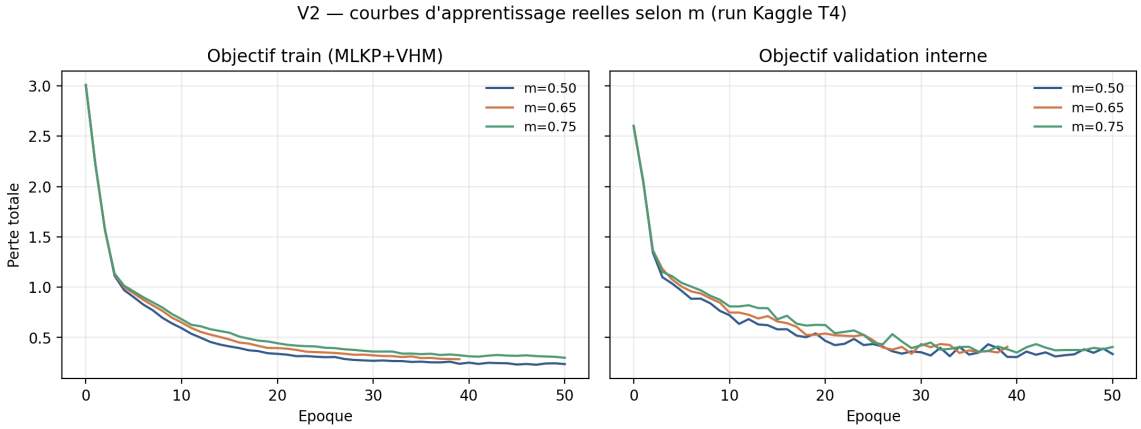


Figure 1.3: Real total MLKP+VHM objective curves (Kaggle run). Train: 0.2356 (m=0.50), 0.2842 (m=0.65, early stop at 40 epochs), 0.2982 (m=0.75); internal validation: 0.3348 / 0.4075 / 0.4058. Minimum loss (m=0.50) does not give the best detection.

The initial drop is shared by all three curves and the train/validation gap stays stable; the triplet m=0.75, g=8, r=0.04 maximises validation F1 (0.8493), with test F1 at 0.8314 (-1.8 pts, healthy generalisation).

Engineer's reading: fast convergence (drop from $\sim$3.0 to $<$1.0 in $\sim$5 epochs), then plateau; stable train/validation gap ($\sim$0.10) with no explosive divergence thanks to NLL capping; effective early stopping for m=0.65. Most importantly, **the best MLKP is not the best detector**: selection uses validation F1, not loss (Figure 1.4).

1.8 Locked-Test Results

Figures 1.5 and 1.6 frame Table 1.2.

V2 detail: Precision 95% CI [0.8186; 0.8346], Recall 95% CI [0.8281; 0.8439], FPR 0.0030, Accuracy 0.9943, **Balanced accuracy 0.9166**, Specificity 0.9970. Validation $\rightarrow$ test gap: 0.8493 $\rightarrow$ 0.8314 (-1.8 pts), healthy generalisation.

Reading: V2 beats V1 by +19.9 pts (P), +3.4 pts (R), +12.7 pts (F1), FPR $\div$2.7. Against the paper: R +5.5 pts, F1 +0.8 pt, P -4.35 pts = **partial exceedance only**.

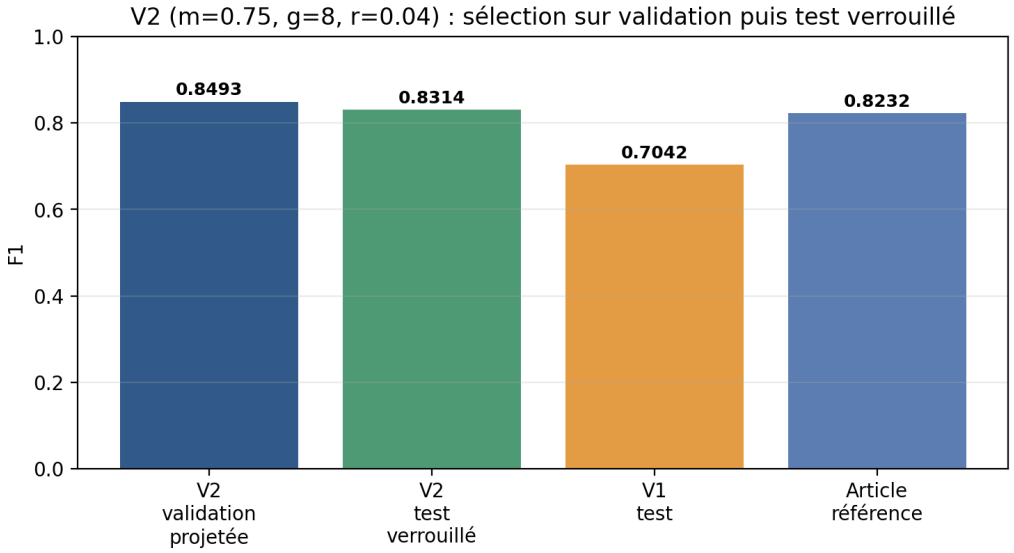


Figure 1.4: Validation-only selection then locked test: projected validation F1 of 0.8493 for triplet m=0.75, g=8, r=0.04 (projected precision 0.8677, recall 0.8317), test F1 0.8314. Test is scored only after freezing.

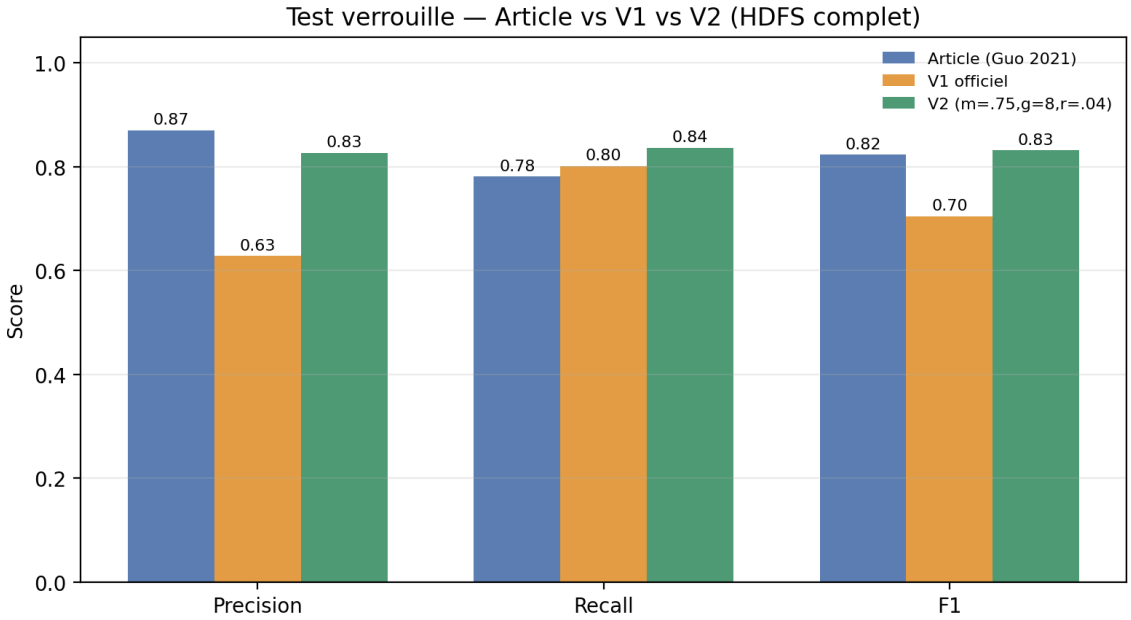


Figure 1.5: Locked full-HDFS test: paper (non-paired reference) vs. official V1 vs. V2.

Table 1.2: Locked-test comparison. Paper: indicative reference.

Version	Precision	Recall	F1	ROC-AUC	PR-AUC
Paper HDFS	0.8702	0.7810	0.8232	—	—
V1 (m=0.65, g=6, r=0)	0.6278	0.8017	0.7042	0.8979	0.6338
V2 (m=0.75, g=8, r=0.04)	**0.8267**	**0.8362**	**0.8314**	**0.9235**	**0.7051**

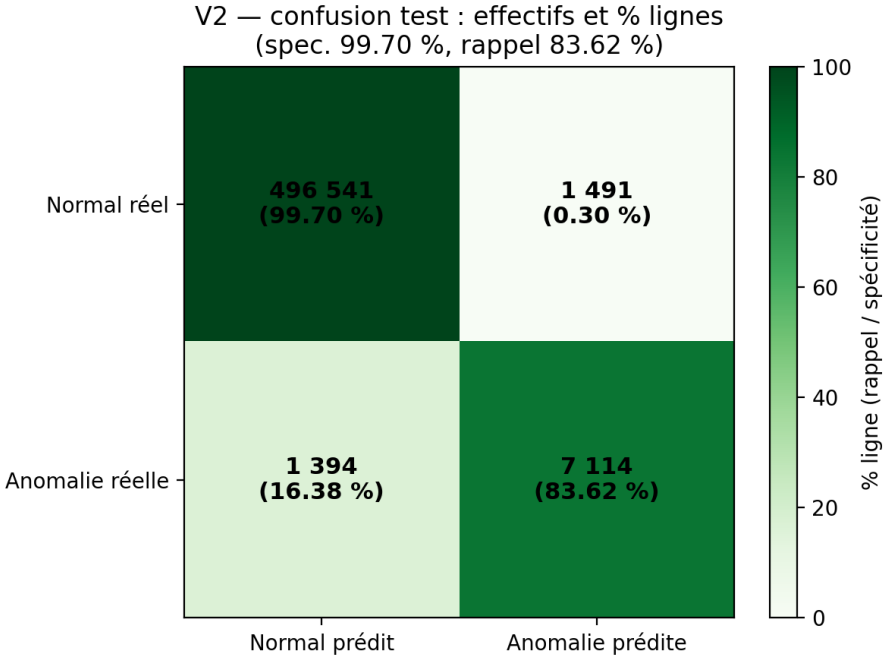


Figure 1.6: V2 matrix in counts and row %: 496,541 correctly classified normals (99.70%) and 7,114 detected anomalies (83.62%); 1,491 FPs (0.30%) and 1,394 FNs (16.38%).

Single seed: insufficient to claim robust superiority (requires 3–5 seeds, mean ± std).

1.9 BGL Generalisation and Multi-Seed Rigor (V3)

Same protocol as V2 (official repository `a845e61`, paper-style scoring, normal-only training, validation selection, locked test), applied to **BGL**: 4.7 million BlueGene/L messages, 5-min sliding windows / 1-min step (average length ∼562 vs. ∼19 for HDFS), seed-12 shuffle, `train_ratio=0.4`. Split: 6,681 normal training sessions; validation 1,010 / 535; test 9,053 / 2,091 (18.76% prevalence). Training-only vocabulary: 185 keys. Grid $m \in \{0.35, 0.50, 0.65\}$ around the official default 0.50; seeds 1234/2026/999 on frozen splits (only initialisation, training and masking vary).

Phase-A selection (seed 1234): m=0.50, g=15, r=0.01 (projected validation F1 0.9119, validation → test gap −0.2 pts). Locked test per seed (Table 1.3, Figure 1.8).

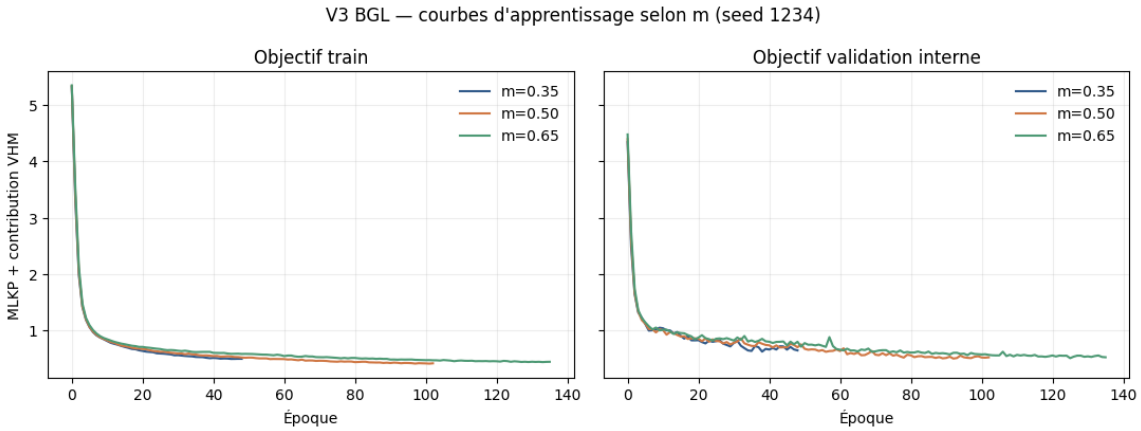


Figure 1.7: V3 BGL — MLKP+VHM curves by m: 49 epochs (m=0.35), 103 (m=0.50), 136 (m=0.65). Effective early stopping, no divergence.

Table 1.3: V3 BGL, locked test per seed; validation-frozen thresholds.

Seed	Precision	Recall	F1	FPR	ROC-AUC
1234	0.9046	0.9158	0.9102	0.0223	0.9595
2026	0.8979	0.9168	0.9072	0.0241	0.9595
999	0.9112	0.9173	0.9142	0.0207	0.9597
Mean ± std (n=3)	0.9045±0.0066	0.9166±0.0007	0.9105±0.0035	0.0223±0.0017	0.9596±0.000
Paper BGL	0.8940	0.9232	0.9083	—	—

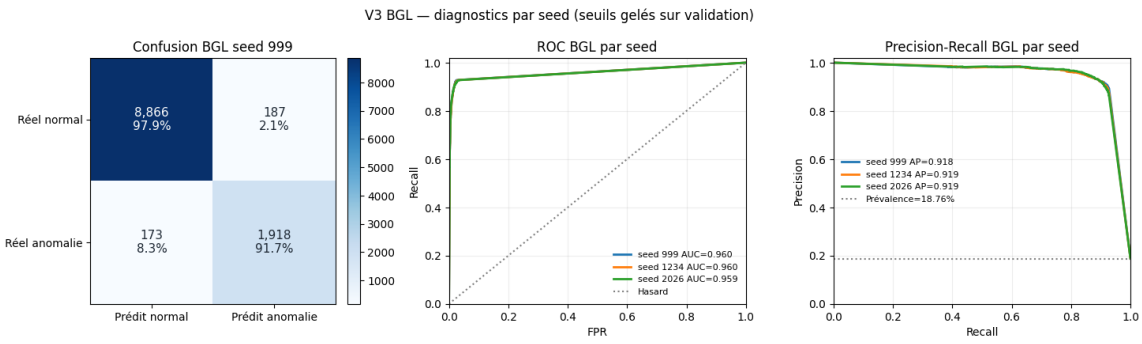


Figure 1.8: V3 BGL — best-seed (999) confusion, per-seed ROC and Precision-Recall (frozen thresholds).

Reading: against the BGL paper (89.40 / 92.32 / 90.83): P +1.05 pts, R −0.66 pts, F1 +0.22 pts, i.e. less than the inter-seed std (±0.35): **a match, not an exceedance**. Multi-seed forbids an abusive claim here — exactly its purpose. Remarkable stability

(recall ±0.0007, ROC ±0.0001).

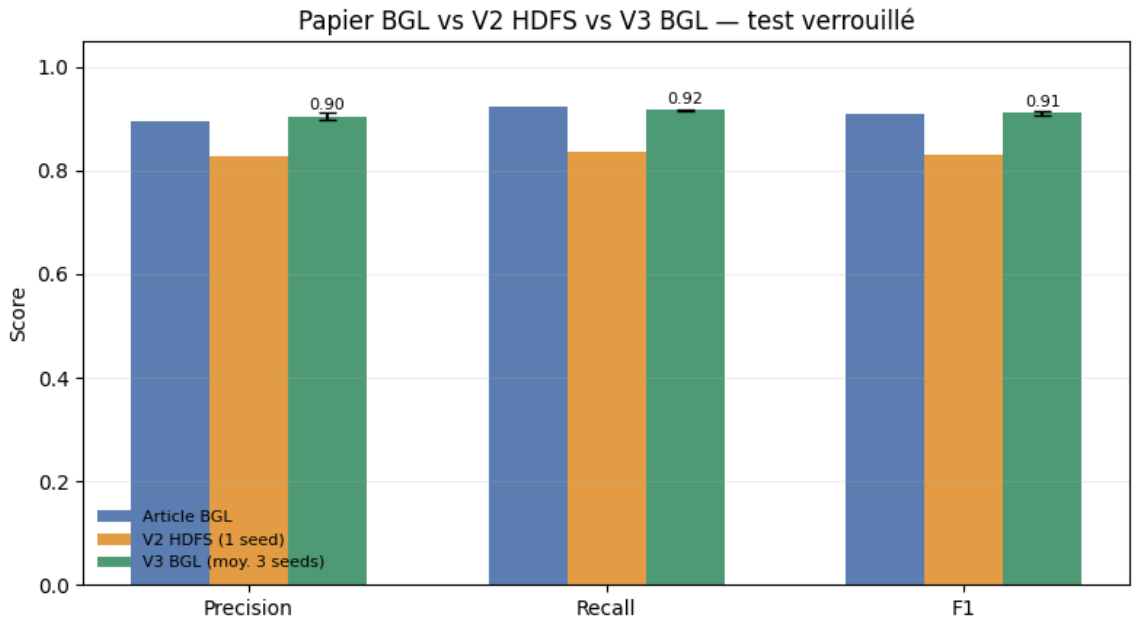


Figure 1.9: Paper BGL vs. V2 HDFS (1 seed) vs. V3 BGL (3-seed mean). Precisions are not comparable across datasets (prevalences 18.76% vs. 1.68%).

Main lesson: the optimal m follows sequence length — HDFS (∼19 tokens) wanted m=0.75, BGL (∼562 tokens) confirms the official m=0.50 default (already masking ∼280 positions per session). The V2/V3 ablation thus forms a unified result rather than two ad-hoc settings. Remaining symmetry: Phase C (extra HDFS seeds on m=0.75) for equal rigor on the HDFS side.

1.10 Inference and Enterprise Applicability

1.10.1 Bundle Inference Contract

Input: list of numeric `EventIds` (Drain output), `session_id`, seed. Output: `anomaly` (score $> r$), `anomaly_score` = out-of-top-g error proportion, `masked_tokens`, `undetected_tokens`, `evidence` (position, `observed_event_id`, template, top-g candidates + templates). Real smoke test: normal (15 masked, 0 errors, score 0.0 → normal); anomalous (18 masked, 1 error, 0.0556 > 0.04 → anomaly), evidence `Event 15` (*Unexpected error trying to delete block..., BlockInfo not found in volumeMap*) outside the 8 expected. Latency: batched DataLoader scoring, no vocabulary recomputation; the SLM is called only on novel signatures.

1.10.2 Worldline (TFI) Integration

Target chain: collectors (Fluent Bit / Logstash) → streaming Drain normalisation → sessionisation (application window: `BlockId` here, `trace_id` / time window in banking production) → LogBERT per application profile (one model per behaviour: payments, CRM, IAM) → SHA256 signature + deduplication + 24-h cache → alert queue (ServiceNow / PagerDuty) with evidence dossier → local SLM in JSON → mandatory human validation (human-in-the-loop). MLOps: model registry, drift monitoring ([UNK] rate, sliding FPR), retraining on validated incidents, GitOps CI/CD.

CommerzBank retrospective: 4,600 lines over 3 components, nested causes. The pipeline brings: simultaneous rupture isolation, immediate infra-vs-application qualification (Worldline responsibility limited to the platform layer), investigation cut from 20–30 min to <5 min of processing (−75 to −83% estimated, to confirm in a blind trial). Under Copartis load (172 M lines/month), measured filtering (∼98.3% discarded, 1.70% kept) + Heaps + cache brings the SLM load down to ∼869 calls/day (working hypothesis).

1.10.3 Enterprise Use Case and Causal-Diagnostic Example (JSON)

Scenario: on the banking scope supervised by Client Delivery 2, an HDFS session is classified **anomalous** by LogBERT V2 (m=0.75, g=8, r=0.04, score 0.0556 > 0.04). The evidence dossier (real excerpt from the bundle smoke test) is forwarded as-is to on-call, then the SLM translates it into an actionable diagnosis without re-deciding.


```
{
  "session_id": "smoke-anomaly",
  "scored": true, "anomaly": true,
  "anomaly_score": 0.0556,
  "m": 0.75, "g": 8, "r": 0.04,
  "masked_tokens": 18, "undetected_tokens": 1,
  "evidence": [{
    "position": 19,
    "observed_event_id": "15",
    "observed_template": "Unexpected error trying to delete
      block blk_<*>. BlockInfo not found in volumeMap.",
    "top_g_candidate_ids": ["5","2","4","10","6","9","3","8"]
  }]
}
```


The Qwen2.5-1.5B-Instruct SLM answers **only** in the constrained schema below (format enforced by the prompt; illustrative content, consistent with the evidence above):

```
{
  "engineer_summary": "Block deletion failure: the
    DataNode cannot find the block in its local table.",
  "logbert_reading": "Event 15 outside the expected top-8
    (block deletion/addition/verification).",
  "probable_causal_chain": "Deletion order -> block
    missing from volumeMap -> unexpected error.",
  "uncertainties": ["Block already deleted?",
    "NameNode/DataNode desync?", "Corrupted disk?"],
  "recommended_actions": [
    {"action": "VERIFY_BLOCK_REPLICATION",
     "why": "confirm block presence",
     "priority": "P1"},
    {"action": "CHECK_DATANODE_VOLUME_MAP",
     "why": "check local consistency",
     "priority": "P2"}
  ],
  "cited_events": ["15"],
  "logbert_score": 0.0556,
  "threshold_r": 0.04,
  "margin_over_threshold": 0.0156
}
```


Verified guardrails: cited events $\in$ LogBERT dossier, fact/hypothesis separation, actions from the finite catalogue, mandatory human validation before any execution.

Deliberate choice: no generative confidence score is requested from the SLM — a self-assessed value from a language model is neither calibrated nor actionable for on-call. The only decision-strength signal is LogBERT's: `anomaly_score` against threshold r (here $0.0556 > 0.04$, margin $+0.0156$) plus the number of out-of-top-g positions. Uncertainty is stated in plain text under `uncertainties`, not as a pseudo-probability. In production, this dossier flows through ServiceNow/PagerDuty with score, top-g evidence and Drain templates attached.

1.10.4 FinOps and Sovereignty

Naive/hybrid ratio $\approx \times 43$. Sovereignty surcharge $\approx +€71$/month ($\times 1.9$): the price of data never leaving the premises (GDPR, PCI-DSS). Dedicated-vs-API break-even $\sim$785 M lines/day. The local SLM (Qwen2.5-1.5B, $\sim$34.4 s/diagnosis, 8.3 GPU-h/day, 35% occupancy) is a compliance choice, not a marginal cost optimisation.

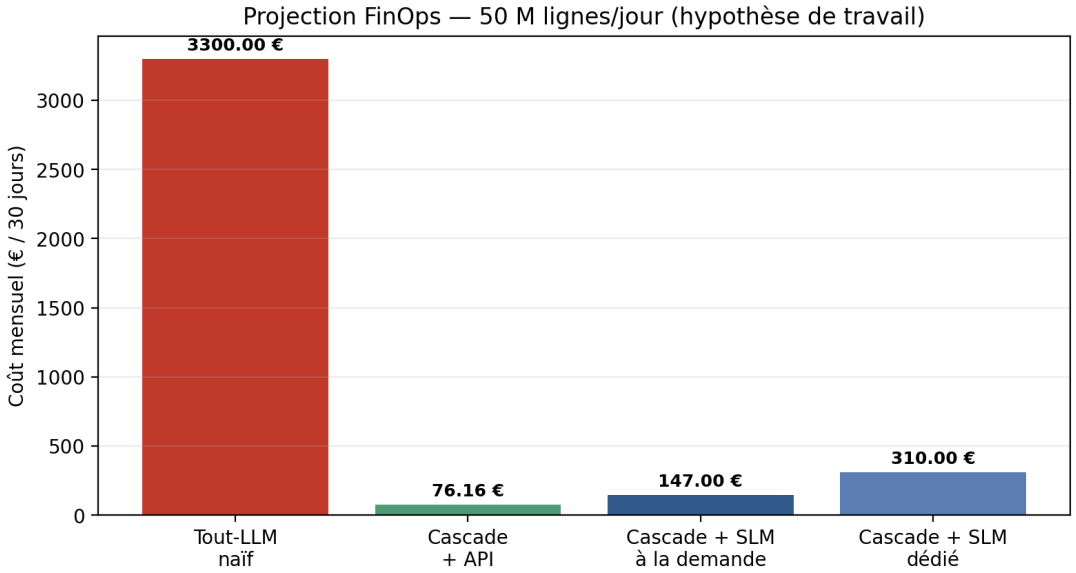


Figure 1.10: Monthly projection at 50 M lines/day: €3,300 (naive all-LLM) vs. €76.16 (cascade+API) vs. €147 (cascade+self-hosted SLM on demand) vs. €310 (dedicated).

1.10.5 Risks, Limitations, Acceptability

Hallucinations neutralised by strict JSON + finite catalogue + deterministic injection; syntactic drift tracked via [UNK]. Limitations: single seed, non-paired paper comparison, NameNode-vs-DataNode attribution bias (fix by injecting cluster topology into context), HDFS-only evaluation to extend to multi-service. Acceptability: evidence dossier attached to every alert, supervised advisor mode.

1.11 Conclusion

Locked V2: **0.8267** / **0.8362** / **0.8314** (1 seed), ROC-AUC 0.9235, PR-AUC 0.7051, FPR 0.30% over 506,540 sessions; V3 BGL: **0.9045** / **0.9166** / **0.9105** (3-seed mean, ±0.0035 in F1), matching the BGL paper (90.83). Hybridisation reconciles detection, constrained explainability and sustainability (÷43, €76–147/month). Partial paper exceedance, to consolidate with multi-seed repeats. Recommendations: per-profile MLOps registry, FinOps-by-design (<1,000 calls/day), human-in-the-loop, blind MTTR measurement.

Bibliography

[1] Vaswani et al. Attention is all you need. NeurIPS, 2017.

[2] Kingma, Welling. Auto-encoding variational Bayes. ICLR, 2013.

[3] Devlin et al. BERT. NAACL, 2018.

[4] He et al. Drain: online log parsing with fixed depth tree. ICWS, 2017.

[5] Guo et al. LogBERT: log anomaly detection via BERT. IJCNN, 2021.

[6] Official LogBERT repository. https://github.com/HelenGuohx/LogBERT.git, commit `a845e61`.

[7] Zhu et al. Tools and benchmarks for automated log parsing. ICSE-SEIP, 2019.

[8] He et al. Loghub: a large collection of system log datasets. ISSRE, 2020.

[9] Du et al. DeepLog: anomaly detection and diagnosis from logs. CCS, 2017.

[10] Xu et al. Online system problem detection by mining patterns of console logs. ICDM, 2009.

[11] Schölkopf et al. Estimating the support of a high-dimensional distribution. Neural Computation, 2001.

[12] Liu et al. Isolation Forest. ICDM, 2008.

[13] Lin et al. Log clustering based problem identification for online service systems. ICSE, 2016.

[14] He et al. Experience report: system log analysis for anomaly detection. ISSRE, 2016.

[15] Meng et al. LogAnomaly: unsupervised detection of sequential and quantitative anomalies. IJCAI, 2019.

[16] Xu et al. Detecting large-scale system problems by mining console logs. SOSP, 2009.

[17] Ndichu et al. Alert fatigue in production supervision, 2026.

[18] Ji et al. Survey of hallucination in NLG. ACM Comp. Surv., 2023.

[19] Qwen Team. Qwen2.5 technical report, 2025.

[20] Meta AI. Llama 3 model card, 2024.

[21] Heaps. Information retrieval. Academic Press, 1978.

[22] Han et al. A survey on AIOps for failure management. ACM Comp. Surv., 2022.

[23] Notaro et al. AIOps: chronology, landscape and challenges. IEEE Trans. Net. Serv. Manag., 2021.

[24] Worldline. Universal registration document 2024. Worldline Investors, revenue €4,632 M, ∼18,000 experts, ∼40 countries. https://investors.worldline.com/.

[25] Worldline. PayTech provider, services to financial institutions and locations. https://worldline.com/.